

Stage-2 Warm-Started Median Penetration NLL

Mie Spray Postprocessing Project

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Audience: Mechanical engineers + IT engineers with basic DL experience.

Goal of this talk:

- Explain what `median_penetration_NLL.ipynb` adds after Stage-1.
- Show how Stage-1 weights are warm-started for heteroscedastic NLL training.
- Summarize the Stage-2 data pipeline, loss, training loop, and saved artifacts.
- Connect the Stage-2 run outputs to later inference notebooks.

1.1 Position in the Curriculum

Stage-1 already learned:

- stable mean penetration trend from fitted penetration curves
- a compatible two-head network output: μ and $\log \sigma^2$

Stage-2 objective in this notebook:

- keep the same MLP interface
- warm-start from `runs_mlp/stage1_median_penetration_20260306_172838`
- replace Stage-1 MSE-centered training by Gaussian NLL
- learn condition-dependent uncertainty instead of only a mean trajectory

1.2 Data Source and Filtering Scope

Input source:

- fitted CSV files under MLP/synthetic_data/*/clean
- each row stores four log-parameters of the fitted spray model

Curve reconstruction model:

$$S(t) = (1 - w(t)) k_{\sqrt{}} \sqrt{t} + w(t) k_{1/4} t^{1/4}, \quad w(t) = \sigma\left(\frac{t - t_0}{s}\right)$$

Current notebook filtering logic:

- compare all candidate curves at 5 ms
- reject penetration outliers with robust z-score threshold $|z| > 3$
- reject nearly flat curves with $|dS/dt| < 10^{-6}$ at compare time
- keep all remaining rows for Stage-2 learning, not only one median row

1.3 Resulting Stage-2 Table

Observed notebook output:

- selected penetration time-series: 21 177
- row-level feature table shape: (21 177, **60**)
- train/val/test split: 14 825 / 3 176 / 3 176

Why this matters:

- Stage-1 used representative supervision to stabilize the mean.
- Stage-2 now exposes the model to a broader trajectory distribution.
- The uncertainty head can finally see condition-dependent spread.

2.1 Physically Consistent Scaling

Nine input features are used:

- `time_norm_0_5ms`
- `tilt_angle_radian_z`, `plumes_z`, `diameter_mm_z`, `injection_duration_us_z`
- `log_injection_pressure_bar_z`, `log_chamber_pressure_bar_z`
- `log_delta_pressure_bar_z`, `control_backpressure_bar_z`

Scaling policy:

- time: fixed min-max over 0 ms to 5 ms
- pressures: transformed in log domain before z-score
- geometry and condition terms: z-score fitted on train split only
- missing defaults: $P_{inj} = 2000$ bar, backpressure = 4 bar

2.2 Row-to-Point Expansion for MLP Training

Each filtered row is decoded to a dense time grid before entering the MLP:

$$\text{one operating condition row} \Rightarrow \{(x_t, y_t)\}_{t=1}^{1024}$$

Implementation choices in the notebook:

- fixed time grid: 1024 points in 0 ms to 5 ms
- static physical features are repeated across all time points
- DataLoader flattens batch tensor from $[B, T, F]$ to $[B \cdot T, F]$

Observed first-batch sanity check:

- feature batch shape: (262 144, 9)
- target batch shape: (262 144, 1)
- all interface checks passed: no NaN/Inf, time in $[0, 1]$, feature dim = 9

3.1 Warm-Started Network

Loaded Stage-1 model:

- input dim = 9, output dim = 2
- hidden dims = [512, 512, 128]
- LayerNorm + GELU + Dropout(0.8)

Why warm-start instead of retraining from scratch:

- Stage-1 already learned a reasonable penetration mean field.
- Stage-2 mainly needs to refine μ and calibrate $\log \sigma^2$.
- This reduces instability when switching from MSE-style training to NLL.

3.2 Stage-2 Objective

$$\mathcal{L}_{\text{stage2}} = \underbrace{\frac{1}{2} \left(\log \sigma^2 + \frac{(y - \mu)^2}{\sigma^2} \right)}_{\text{heteroscedastic Gaussian NLL}} + \lambda_{d1} \phi(d\mu/dt < 0) + \lambda_{d2} \phi(d^2\mu/dt^2 > 0)$$

Notebook details:

- predicted log_var is clamped to [-10, 6]
- $d1$ penalty enforces non-decreasing penetration
- $d2$ penalty is softly activated after 0.5 ms
- current Stage-2 weights: $\lambda_{d1} = 5 \times 10^{-5}$, $\lambda_{d2} = 5 \times 10^{-5}$

3.3 Why NLL Is the Right Upgrade

If the model predicts too little variance:

$$\frac{(y - \mu)^2}{\sigma^2}$$

becomes very large whenever the trajectory spread is high.

If the model predicts too much variance:

$$\log \sigma^2$$

penalizes that over-cautious choice.

Engineering interpretation:

- stable operating conditions should lead to low predicted variance
- noisy operating conditions should lead to larger predicted variance
- the uncertainty output becomes a meaningful regression target, not just a placeholder head

4.1 Training Loop and Saved Artifacts

Optimizer and schedule inherited from config:

- AdamW, $\text{LR} = 4 \times 10^{-3}$, $\text{weight decay} = 2 \times 10^{-4}$
- gradient clipping = 1.0
- early stopping patience = 50, $\text{min-delta} = 10^{-5}$
- up to 750 epochs

Artifacts written by the notebook:

- `iteration_loss.csv`
- `epoch_loss.csv`
- `loss_curves.png`
- `model checkpoint + scaler_state.json + train_config_used.json`

4.2 Example Run Summary

Observed run directory:

- runs_mlp/stage2_NLL_penetration_20260317_194155

Epoch-loss summary from epoch_loss.csv:

- best validation loss $\approx$ **2.449** at epoch **570**
- training continued to epoch **620**
- final train loss remained above val loss, consistent with strong regularization

Practical reasons for train loss $>$ val loss:

- Dropout(0.8) only perturbs training forward passes
- derivative penalties are active in training objective
- train split may contain harder conditions than validation

Takeaway

- `median_penetration_NLL.ipynb` upgrades Stage-1 mean learning into Stage-2 uncertainty-aware training.
- The notebook keeps the same 9-feature interface and 2-head MLP, but warm-starts from the Stage-1 checkpoint.
- NLL plus mild shape penalties is the main mechanism for learning heteroscedastic penetration behavior.
- The saved Stage-2 artifacts directly support later notebooks such as `toy_inference_from_run.ipynb`.